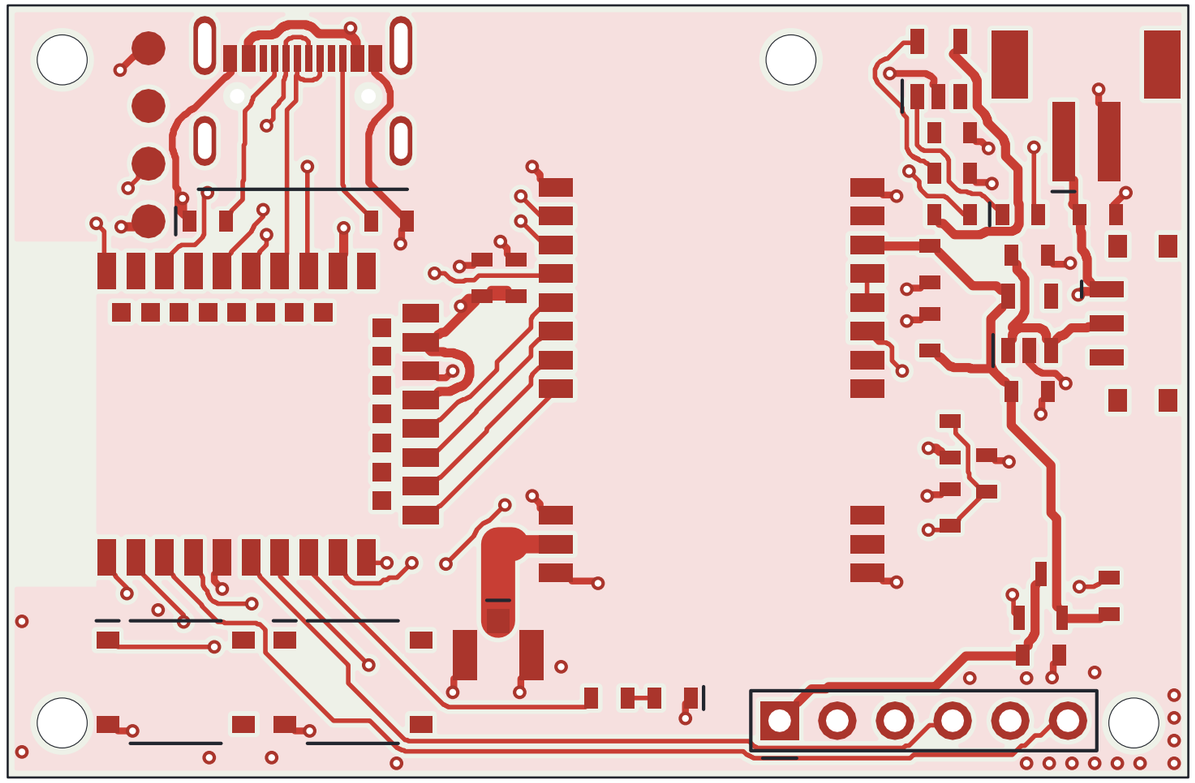
LoRa Mesh Node Mini

nRF52840 (E73) + SX1262 (E22, EU868), USB-C charging, 1S LiPo, GPS header

REVISION	DATE	AUTHOR	COMPANY
A	2026-11-02	SolidCraft example	–



Top side · board 52.0 × 34.0 mm, 2 copper layers, 33 parts fitted on both sides

Contents

2 Power
3 USB
4 MCU
5 LoRa
6 GPS
7 UI
8 Block diagram
9 Parts list (1 of 2)
10 Parts list (2 of 2)
11 Board top
12 Board bottom (seen from the bottom)
13 Assembly top
14 Assembly bottom (seen from the bottom)
15 Layer stack-up

Revisions

REV	DATE	BY	NOTE
A	2026-11-02	SolidCraft example	–

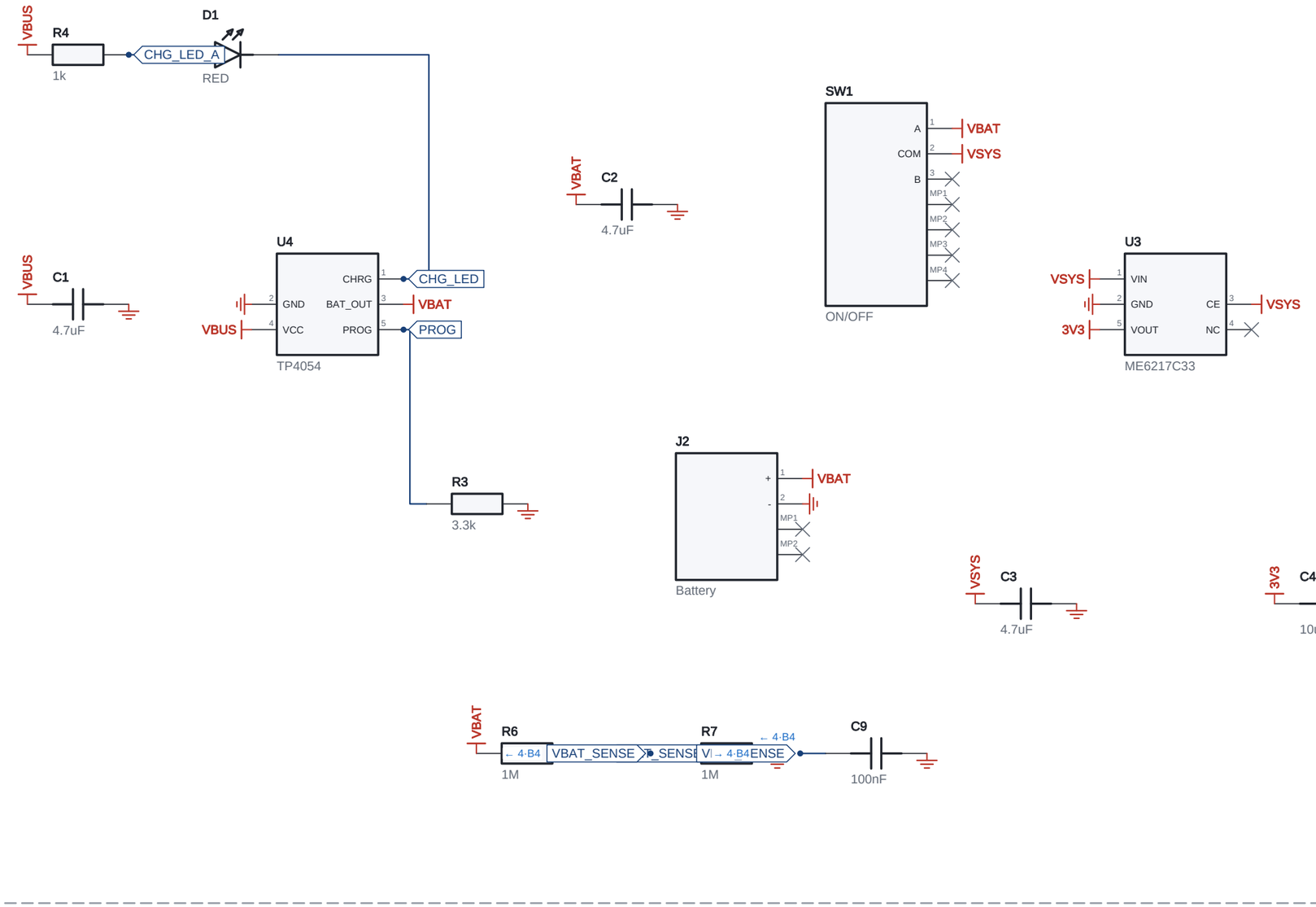
LoRa Mesh Node Mini — SolidCraft

Cover — nRF52840 (E73) + SX1262 (E22, EU868), USB-C charging, 1…

PAGE	REVISION	DATE	SCALE
1 of 15	A	2026-11-02	–

AUTHOR · COMPANY	PAPER	FILE
SolidCraft example	A3 landscape	phase1.sc

Power



LoRa Mesh Node Mini

SolidCraft

Power

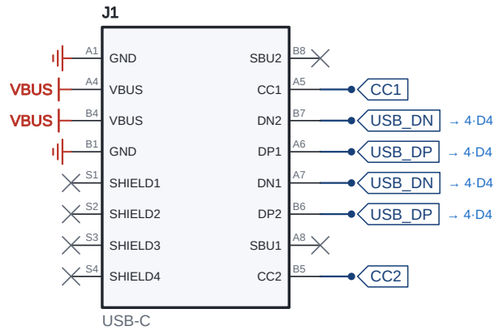
nRF52840 (E73) + SX1262 (E22, EU868), USB-C charging, 1…

PAGE	REVISION	DATE	SCALE
2 of 15	A	2026-11-02	2:1

AUTHOR · COMPANY	PAPER	FILE
SolidCraft example	A3 landscape	phase1.sc

USB





LoRa Mesh Node Mini

SolidCraft

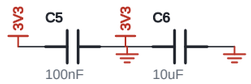
USB

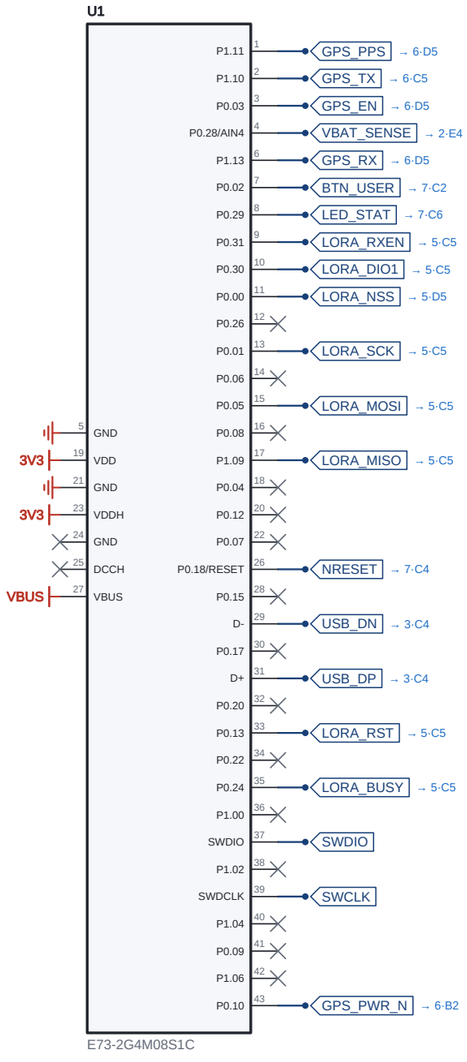
nRF52840 (E73) + SX1262 (E22, EU868), USB-C charging, 1…

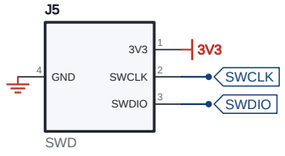
PAGE	REVISION	DATE	SCALE
3 of 15	A	2026-11-02	2:1

AUTHOR · COMPANY	PAPER	FILE
SolidCraft example	A3 landscape	phase1.sc

MCU







LoRa Mesh Node Mini

SolidCraft

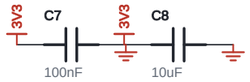
MCU

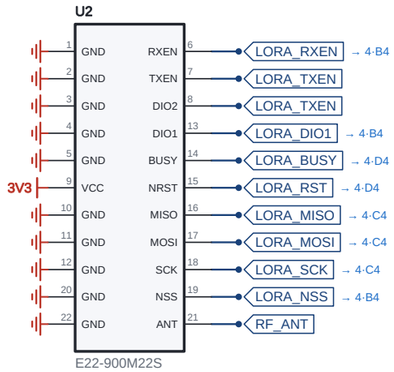
nRF52840 (E73) + SX1262 (E22, EU868), USB-C charging, 1…

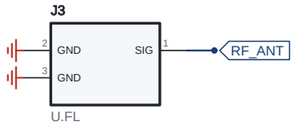
PAGE	REVISION	DATE	SCALE
4 of 15	A	2026-11-02	1.75:1

AUTHOR · COMPANY	PAPER	FILE
SolidCraft example	A3 landscape	phase1.sc

LoRa







LoRa Mesh Node Mini

SolidCraft

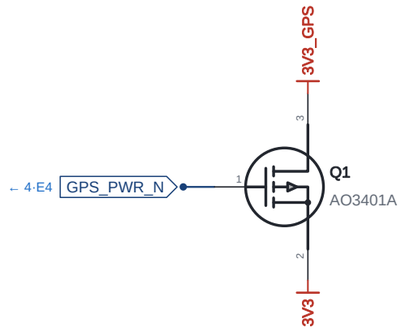
LoRa

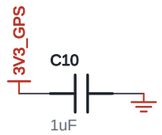
nRF52840 (E73) + SX1262 (E22, EU868), USB-C charging, 1…

PAGE	REVISION	DATE	SCALE
5 of 15	A	2026-11-02	1.75:1

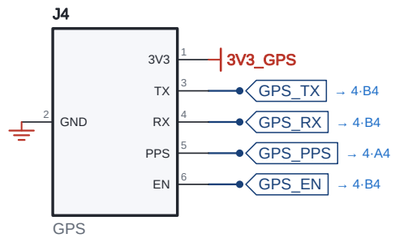
AUTHOR · COMPANY	PAPER	FILE
SolidCraft example	A3 landscape	phase1.sc

GPS









LoRa Mesh Node Mini

SolidCraft

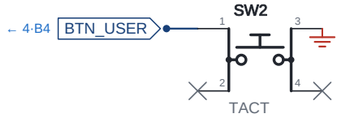
GPS

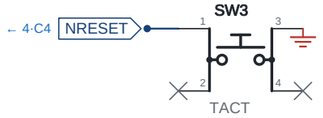
nRF52840 (E73) + SX1262 (E22, EU868), USB-C charging, 1…

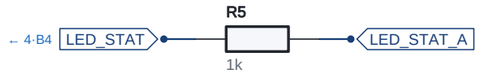
PAGE	REVISION	DATE	SCALE
6 of 15	A	2026-11-02	2:1

AUTHOR · COMPANY	PAPER	FILE
SolidCraft example	A3 landscape	phase1.sc

UI









LoRa Mesh Node Mini

SolidCraft

UI

nRF52840 (E73) + SX1262 (E22, EU868), USB-C charging, 1…

PAGE	REVISION	DATE	SCALE
7 of 15	A	2026-11-02	2:1

AUTHOR · COMPANY	PAPER	FILE
SolidCraft example	A3 landscape	phase1.sc

Signals (bus: shared name prefix, count)

Supply rails: → made in this frame, ← from the page of the frame that makes it; dashed: (detect), (sense) or (pull-up), a signal or reference, not what powers the frame

Power tree (bottom left): feeds detect, sense or pull-up: a signal or reference, not what powers the frame

MCU
U1 E73-2G4M08S1C
J5 SWD-PADS
4 parts · p. 4

3V3 ← p. 2
VBUS (detect) ← p. 3

LoRa
U2 E22-900M22S
J3 U.FL
4 parts · p. 5

3V3 ← p. 2

GPS
J4 GPS-HDR
Q1 AO3401A
4 parts · p. 6

3V3 ← p. 2

LORA (8)

GPS (5)

VBAT_SENSE

Power
SW1 SLIDE
U3 ME6217C33M5G
U4 TP4054
J2 JST-PH-2
14 parts · p. 2

3V3 → 3 frames
VBUS ← p. 3

USB
J1 USB-C
3 parts · p. 3

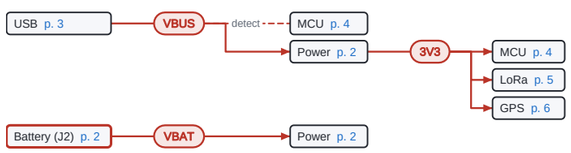
VBUS → p. 2, 4

USB (2)

UI
SW2 TACT
SW3 TACT
4 parts · p. 7

BTN_USER, LED_STAT, NRESET

Power tree



LoRa Mesh Node Mini

SolidCraft

Block diagram

nRF52840 (E73) + SX1262 (E22, EU868), USB-C charging, 1…

PAGE	REVISION	DATE	SCALE
8 of 15	A	2026-11-02	–

AUTHOR · COMPANY	PAPER	FILE
SolidCraft example	A3 landscape	phase1.sc

Parts list

33 parts to fit, in 23 lines · 0 not fitted

Qty	Designators	Value	Part	Footprint	MPN · Manufacturer	Description
3	C1–C3	4.7uF	C-4u7	0603	CL10A475KO8NNNC Samsung	
3	C4, C6, C8	10uF	C-10u	0603	CL10A106KP8NNNC Samsung	
3	C5, C7, C9	100nF	C-100n	0603	CC0603KRX7R9BB104 YAGEO	
1	C10	1uF	C-1u	0603	CL10A105KB8NNNC Samsung	
1	D1	RED	LED-RED	LED0603	KT-0603R Hubei KENTO	
1	D2	GREEN	LED-GREEN	LED0603	19-217/GHC-YR1S2/3T Everlight	
1	J1	USB-C	USB-C	USB-C-16P	TYPE-C-31-M-12 Korean Hroparts	
1	J2	Battery	JST-PH-2	JST-PH-2-SMD	S2B-PH-SM4-TB(LF)(SN) JST	
1	J3	U.FL	U.FL	U.FL	U.FL-R-SMT-1(10) Hirose	
1	J4	GPS	GPS-HDR	HDR-1x6		
1	J5	SWD	SWD-PADS	SWD-PADS-4		
1	Q1	AO3401A	AO3401A	SOT-23	AO3401A AOS	
2	R1, R2	5.1k	R-5k1	0603	0603WAF5101T5E UNI-ROYAL	
1	R3	3.3k	R-3k3	0603	0603WAF3301T5E UNI-ROYAL	
2	R4, R5	1k	R-1k	0603	0603WAF1001T5E UNI-ROYAL	
2	R6, R7	1M	R-1M	0603	0603WAF1004T5E UNI-ROYAL	
1	R8	100k	R-100k	0603	0603WAF1003T5E UNI-ROYAL	
1	SW1	ON/OFF	SLIDE	SLIDE-SPDT	MSK12C02 SHOU HAN	
2	SW2, SW3	TACT	TACT	TACT-5.1	TS-1187A-B-A-B XKB	
1	U1	E73-2G4M08S1C	E73-2G4M08S1C	E73-2G4M08S1C	E73-2G4M08S1C Ebyte	
1	U2	E22-900M22S	E22-900M22S	E22-900M22S	E22-900M22S Ebyte	
1	U3	ME6217C33	ME6217C33M5G	SOT-23-5	ME6217C33M5G Microne	

Continued on the next page

LoRa Mesh Node Mini — SolidCraft

Parts list (1 of 2)

nRF52840 (E73) + SX1262 (E22, EU868), USB-C charging, 1…

PAGE	REVISION	DATE	SCALE
9 of 15	A	2026-11-02	–

AUTHOR · COMPANY	PAPER	FILE
SolidCraft example	A3 landscape	phase1.sc

Parts list

33 parts to fit, in 23 lines · 0 not fitted

Qty	Designators	Value	Part	Footprint	MPN · Manufacturer	Description
1	U4	TP4054	TP4054	SOT-23-5	TP4054 UMW	

LoRa Mesh Node Mini

SolidCraft

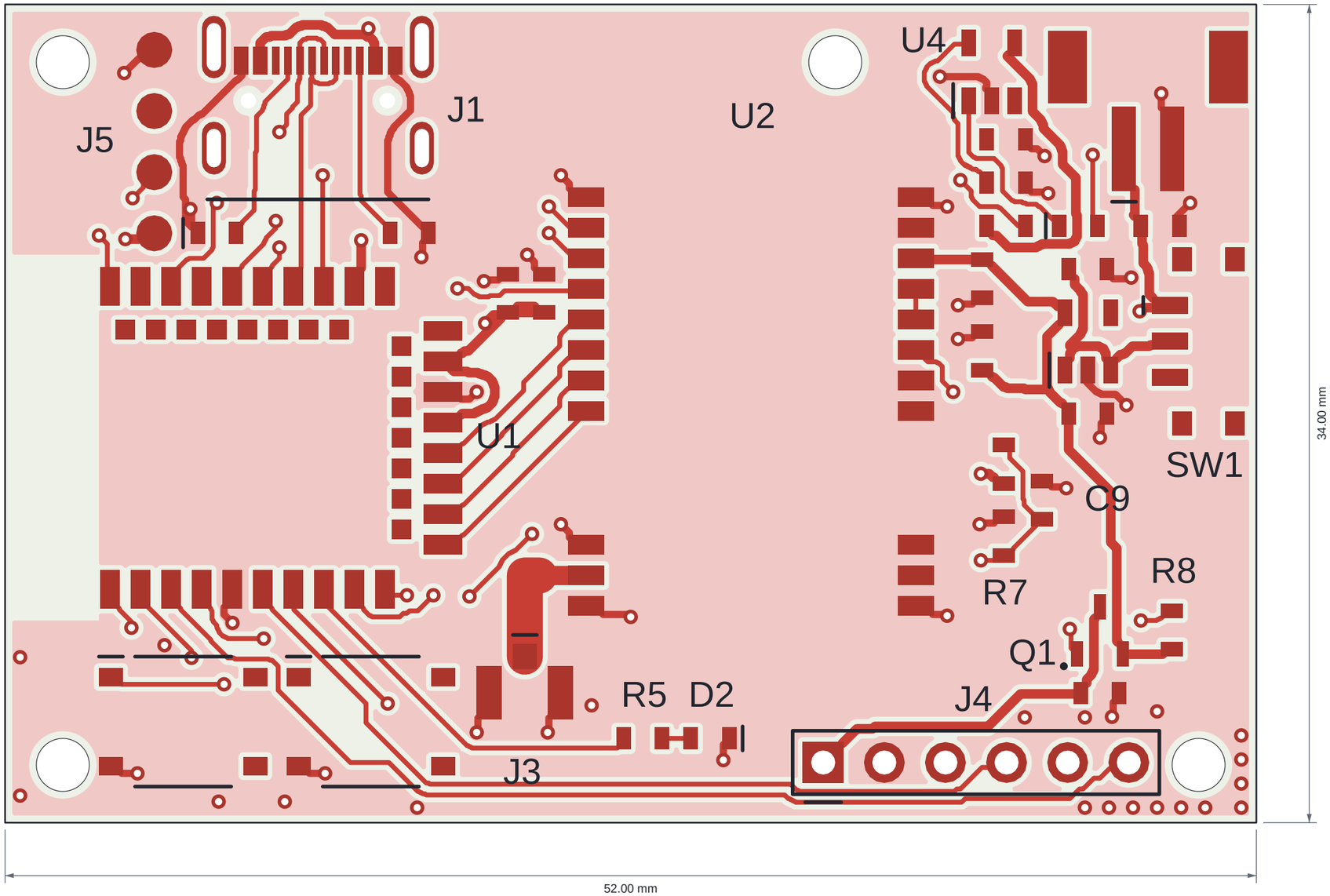
Parts list (2 of 2)

nRF52840 (E73) + SX1262 (E22, EU868), USB-C charging, 1…

PAGE	REVISION	DATE	SCALE
10 of 15	A	2026-11-02	–

AUTHOR · COMPANY	PAPER	FILE
SolidCraft example	A3 landscape	phase1.sc

Top copper, pours, pads and vias · top silkscreen · board outline



LoRa Mesh Node Mini

SolidCraft

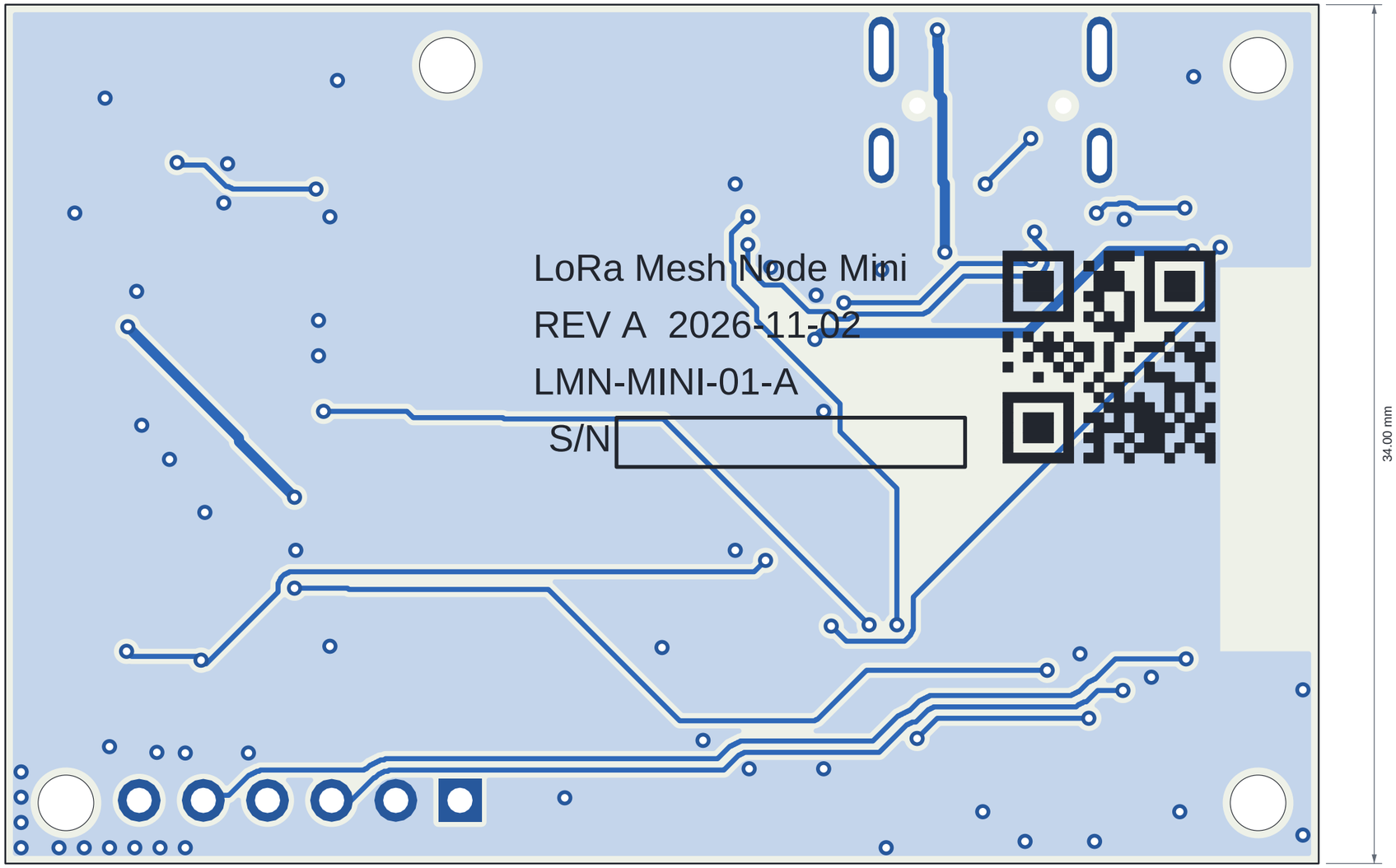
Board top

nRF52840 (E73) + SX1262 (E22, EU868), USB-C charging, 1…

PAGE	REVISION	DATE	SCALE
11 of 15	A	2026-11-02	5:1

AUTHOR · COMPANY	PAPER	FILE
SolidCraft example	A3 landscape	phase1.sc

Seen from the bottom (mirrored) · bottom copper · bottom silkscreen



LoRa Mesh Node Mini

SolidCraft

Board bottom (seen from the bottom)

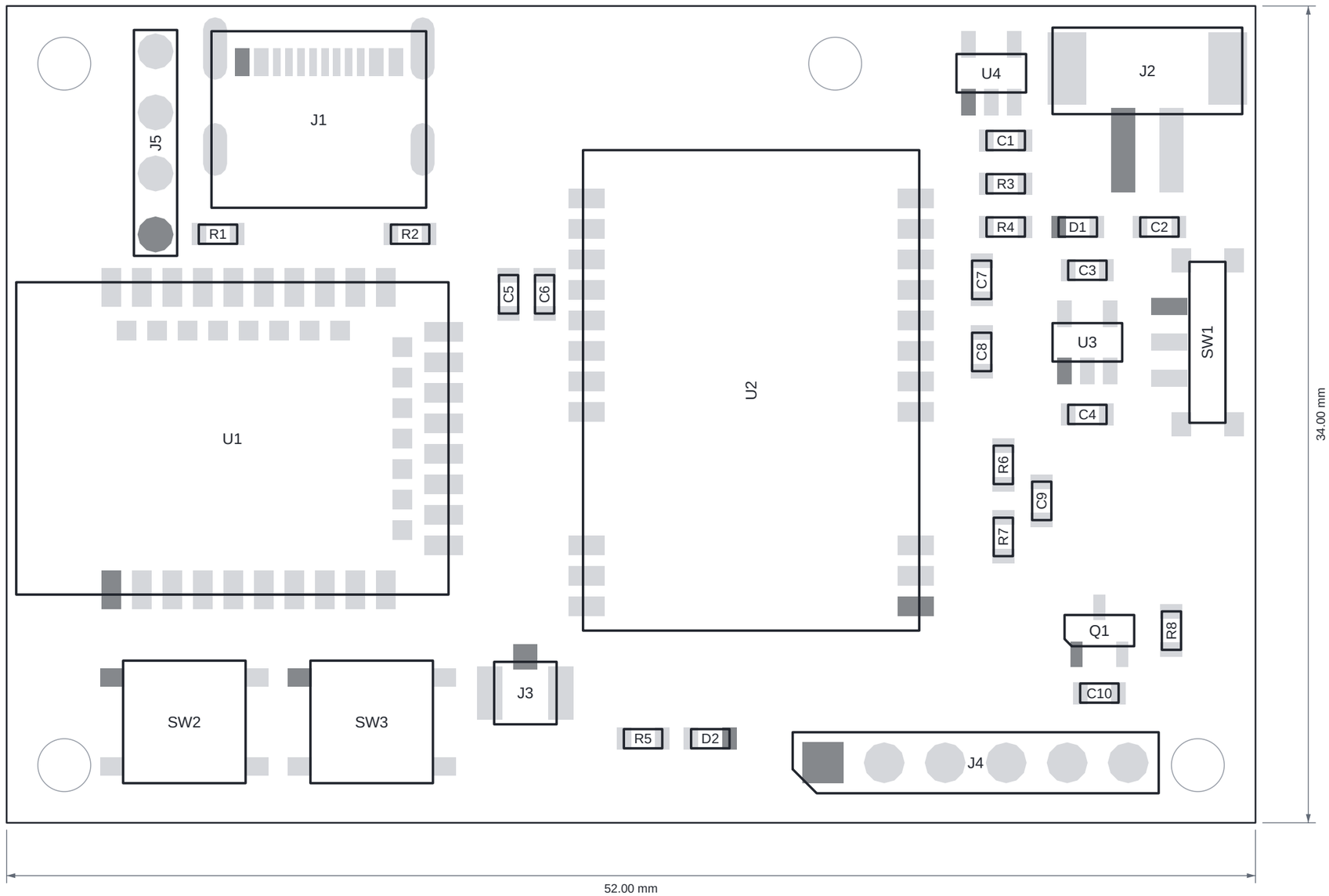
nRF52840 (E73) + SX1262 (E22, EU868), USB-C charging, 1…

PAGE	REVISION	DATE	SCALE
12 of 15	A	2026-11-02	5:1

AUTHOR · COMPANY	PAPER	FILE
SolidCraft example	A3 landscape	phase1.sc

Parts on the top: outlines and designators

33 parts on this side · pad 1 and cathodes darker · values and part numbers: parts list, page 9



LoRa Mesh Node Mini

SolidCraft

Assembly top

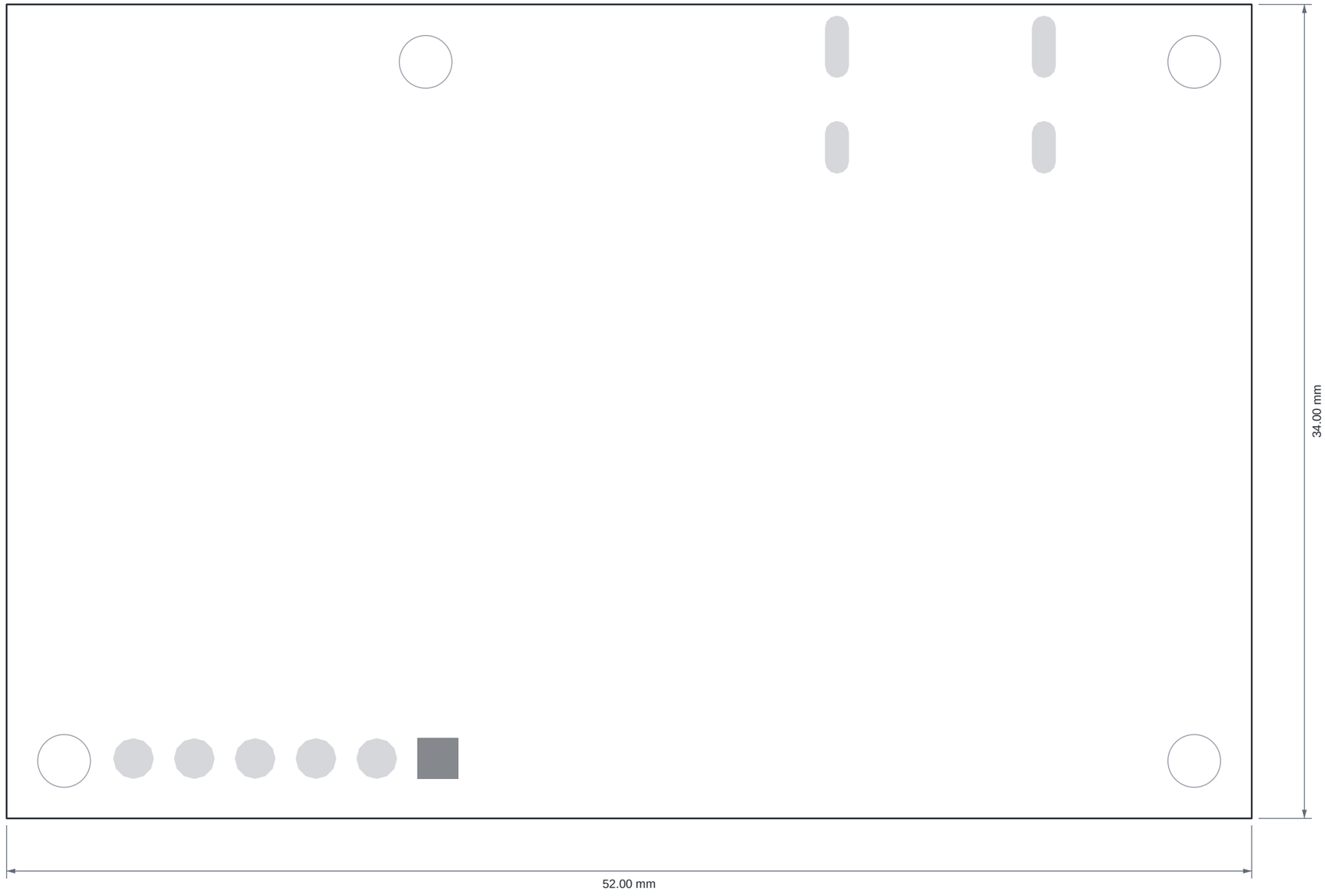
nRF52840 (E73) + SX1262 (E22, EU868), USB-C charging, 1…

PAGE	REVISION	DATE	SCALE
13 of 15	A	2026-11-02	5:1

AUTHOR · COMPANY	PAPER	FILE
SolidCraft example	A3 landscape	phase1.sc

Parts on the bottom, seen from the bottom (mirrored): outlines and designators

0 parts on this side · pad 1 and cathodes darker · values and part numbers: parts list, page 9



LoRa Mesh Node Mini

SolidCraft

Assembly bottom (seen from the bottom)

nRF52840 (E73) + SX1262 (E22, EU868), USB-C charging, 1…

PAGE	REVISION	DATE	SCALE
14 of 15	A	2026-11-02	5:1

AUTHOR · COMPANY	PAPER	FILE
SolidCraft example	A3 landscape	phase1.sc

Layer stack-up

2 copper layers · 1.600 mm finished (1.60 mm board)

Layer	Kind	Thickness	Copper	εr	tan δ
Top	Copper	0.035 mm	1.0 oz		
Core 1	Core	1.530 mm		4.30	
Bottom	Copper	0.035 mm	1.0 oz		

Net classes on the top layer

Class	Width	Gap	Clearance	Impedance
Power	0.400 mm	–	0.200 mm	118 Ω

Impedance: IPC-2141 microstrip (an edge-coupled pair for a class with one) over the nearest plane. A guide: build to the fab's own figures.

Cross-section (not to scale)



LoRa Mesh Node Mini — SolidCraft

Layer stack-up — nRF52840 (E73) + SX1262 (E22, EU868), USB-C charging, 1…

PAGE	REVISION	DATE	SCALE
15 of 15	A	2026-11-02	–

AUTHOR · COMPANY	PAPER	FILE
SolidCraft example	A3 landscape	phase1.sc